Edge-Deployed Infrared (IR) Detection of Occluded Wildlife During Agricultural Harvesting

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Abstract

Agricultural harvesting causes significant wildlife mortality when animals hide within standing crops [1]. Infrared (IR) imaging can detect body-heat signatures through partial vegetation, yet existing computer-vision models target overhead drone or roadside views rather than ground-level machinery. This work proposes a machine-learning pipeline for harvester-mounted IR cameras using a structurally modified YOLO detector with attention mechanisms and occlusion-tolerant loss functions to localize fragmented heat signatures under thick crop cover. A ground-level IR dataset spanning light, medium, and heavy crop densities will be used to characterize detection boundaries, informing future harvester-mounted wildlife-protection systems.

1. Introduction and Motivation

Harvesting machinery routinely injures or kills fawns, Nilgai, and ground-nesting birds that instinctively freeze in standing crops rather than flee, causing thousands of deaths each spring and early-summer harvest [1], [3]. Visual camouflage limits detection by human operators and conventional RGB sensors [8], whereas an IR camera registers an animal's own body heat against cooler vegetation without external illumination, remaining effective under low-light conditions such as dawn, when thermal contrast is greatest [1], [8]. Single-stage detectors, particularly YOLO variants, are effective for IR object detection [2], [7], but existing agricultural deployments target aerial surveying or static perimeter monitoring rather than the ground-level, forward-motion, severely-occluded view characteristic of an active harvester moving through dense standing crop, where the sensor geometry and occlusion pattern differ fundamentally from an overhead drone view. This work investigates an edge-deployed IR vision pipeline specifically adapted for harvester integration, targeting sub-second operator alerts before physical contact occurs, and aims to empirically map the boundary between crop densities where automated detection remains reliable and those where it does not.

2. Problem Definition

We address the real-time detection of animals occluded by standing crops from a ground-level, forward-facing harvester perspective - a substantially harder occlusion and viewing-angle regime than the overhead aerial case addressed in prior work, where the sensor typically looks down through a short vertical column of vegetation rather than horizontally through many meters of standing crop.

- Input & Output: Forward-facing IR video feed → real-time bounding-box localization with confidence scores, triggering operator alerts and, in a future iteration, an automatic slow-down signal.
- Task Formulation: Object detection under severe, structured vegetative occlusion, where only fragments of an animal's IR signature may be visible in a given frame.
- Operational Constraints: Sub-100ms inference latency; fully localized edge computation without cloud reliance; robustness to vibration, dust, and variable ambient temperature.
- Research Question: How can single-stage detectors be structurally modified to localize fragmented, occluded IR signatures from a ground-level harvester vantage point, and where, as a function of crop density, does that reliability break down?

3. Related Work and Research Gap

Aerial scouting systems using modified Faster R-CNN [4] or YOLO-based detectors [7] report detection probability falling below 25% under dense canopy [9], [10], showing overhead sensing alone is insufficient once canopy closure is significant. Perimeter-intrusion systems [5] monitor static field borders and do not address animals already within standing crop at harvest time, while roadside-safety detectors [6], [8] operate in open, largely unoccluded scenes. Recent lightweight architectures, including ALSS-YOLO with adaptive channel splitting [2] and MBAF-MSCNet with multi-branch attention fusion [11], improve detection of small, low-contrast IR targets on edge hardware, but have not been adapted to the ground-level geometry, forward motion, and progressive crop-density occlusion of active harvesting - the gap this project addresses.

4. Proposed Technical Approach and Methodology

We propose an end-to-end edge pipeline combining custom ground-level data collection with targeted model optimizations, structured as four stages: forward-facing IR capture, frame preprocessing, an edge-deployed structurally modified YOLO detector, and thresholded operator alerts (Fig. 1). “Proposed Technical Approach” and “Methodology” refer to the same underlying pipeline throughout this report, with three internal modifications: (i) a lightweight Coordinate Attention backbone that prioritizes localized IR anomalies over repetitive canopy noise; (ii) a feature-preserving detection neck that retains high-resolution spatial maps so small or heavily masked targets survive downsampling; and (iii) an angle-sensitive SIoU loss that improves bounding-box regression when object boundaries are partially obscured by vegetation. A custom ground-level IR dataset of live animals and heated surrogates, staged across low, medium, and high crop densities, will support density-stratified occlusion analysis. The IR camera used for capture is not yet finalized: *the final IR camera specifications will be determined during the hardware selection and validation stage.*

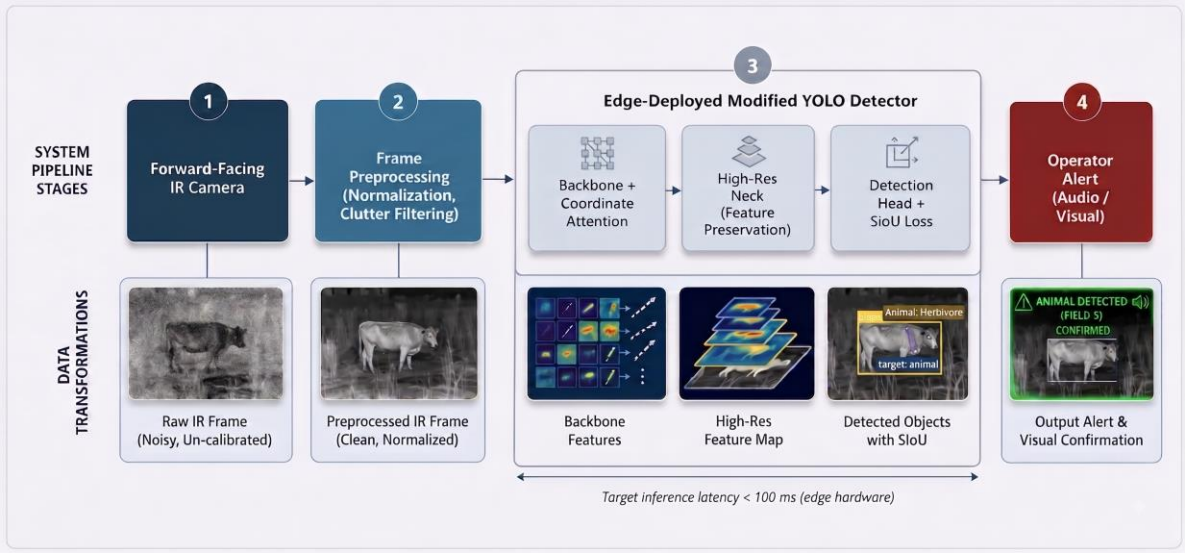


Fig. 1. Proposed harvester-integrated IR detection pipeline. Camera specifications shown are illustrative targets, to be finalized during hardware selection.

5. Current Status and Experimental Plan

The project is in progress: ground-level IR data collection under controlled staging conditions is underway, ensuring consistent lighting, distance, and crop-density labelling across samples to support a clean per-density ablation later. In parallel, we are implementing the attention, feature-preservation, and SIoU modifications on a YOLOv8/v11 baseline, and setting up the edge-hardware deployment environment used for the latency benchmarks below. Present results are methodological - establishing the data-collection protocol, pipeline architecture, and evaluation framework - rather than final detection figures, which are expected once dataset annotation and initial training are complete.

Models will be deployed on representative edge hardware paired with an IR sensor and evaluated separately across low, medium, and high crop-density conditions to isolate the effect of occlusion severity on detection performance. Evaluation will prioritize mAP@0.5, Recall (treated as the primary metric, since false negatives are safety-critical in this application), and inference latency (target <100 ms). Ablation studies will compare the modified YOLO architecture against standard YOLOv8/v11 baselines, and will separately quantify the incremental contribution of the attention module, the feature-preservation neck, and the SIoU loss, individually and in combination.

6. Expected Contribution and Broader Relevance

This work aims to contribute (i) an edge-deployable, occlusion-aware IR detection system for harvester integration, and (ii) an empirical characterization of IR detection limits under dense agricultural occlusion, complementing prior aerial-IR findings that report accuracy below 25% under dense canopy [9], [10]. Immediate next steps include completing dataset annotation, training the modified YOLO variants, conducting ablation experiments across the three crop-density regimes, and field validation alongside an operating harvester where feasible. Beyond this application, the approach has broader relevance to precision-agriculture safety systems and edge-ML detection under severe structured occlusion, and may inform future sensor-fusion approaches combining IR imaging with acoustic or LiDAR cues.

References

[1] P. Christiansen, K. A. Steen, R. N. Jørgensen, and H. Karstoft, "Automated detection and recognition of wildlife using thermal cameras," Sensors, vol. 14, no. 8, pp. 13778–13793, 2014.

[2] A. He, X. Li, X. Wu, C. Su, J. Chen, S. Xu, and X. Guo, "ALSS-YOLO: An adaptive lightweight channel split and shuffling network for TIR wildlife detection in UAV imagery," IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens., 2024.

[3] A. Jarnemo, "Roe deer Capreolus capreolus fawns and mowing - mortality rates and countermeasures," Wildlife Biology, vol. 8, no. 3, pp. 211–218, 2002.

[4] H. Lyu, F. Qiu, L. An, D. Stow, R. Lewison, and E. Bohnett, "Advancing wild deer monitoring through UAV thermal imaging and modified Faster R-CNN," Sensing and Imaging, vol. 25, art. 50, 2024.

[5] N. Mamat, M. F. Othman, and F. Yakub, "Animal intrusion detection in farming area using YOLOv5 approach," in 2022 22nd Int. Conf. Control, Automation and Systems (ICCAS), 2022.

[6] Y. Munian, A. Martinez-Molina, D. Miserlis, H. Hernandez, and M. Alamaniotis, "Intelligent system utilizing HOG and CNN for thermal image-based detection of wild animals in nocturnal periods for vehicle safety," Applied Artificial Intelligence, vol. 36, no. 1, p. 2031825, 2022.

[7] P. Povlsen, D. Bruhn, P. Durdevic, D. O. Arroyo, and C. Pertoldi, "Using YOLO object detection to identify hare and roe deer in thermal aerial video footage," Drones, vol. 8, no. 1, p. 2, 2024.

[8] H. Puppala, W. Sarasua, S. Biyaguda, F. Farzinpour, and M. Chowdhury, "Real-time deer detection and warning in connected vehicles via thermal sensing and deep learning," 2025.

[9] J. Rietz et al., "Drone-based thermal imaging in the detection of wildlife carcasses and disease management," Transboundary and Emerging Diseases, vol. 2023, p. 5517000, 2023.

[10] M. Winsen, S. Denman, E. Corcoran, and G. Hamilton, "Automated detection of koalas with deep learning ensembles," Remote Sensing, vol. 14, no. 10, p. 2432, 2022.

[11] Y. Wu, X. Liu, J. Hao, Z. Xing, H. Xi, W. Sun, and Q. Yao, "MBAF-MSCNet: Multi-branch fusion for multi-scale context in infrared wildlife detection," IEEE Access, vol. 13, pp. 165830–165843, 2025.